

USEPA
DMR-QA 35

NPDES PERMITTEE
DATA REPORT FORM



Weyford Co
Mentzel
ERA

A Waters Company

USEPA NPDES MI0020257

Permit #:

Permit Ext:

Permittee Name: Cadillac WWTP

Facility Address: 1121 Pleitt Rd.

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

Fax Number: (231) 775-2876

E-mail address: lab@cadillac-mi.net

For DMRQA-35, conducted in 2015, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Yes ☒ No ☐

Submitted Complete and Accurate Data
by July 10, 2015

Yes ☒ No ☐

Received a Graded Report by July
31, 2015

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative

(as per 40 C.F.R. Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) Permit. Neither I nor any of my subordinates compared our results from independent analyses conducted by us or any other laboratory before we reported our results to the USEPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of
Certifying Official: Amy Vail

Title: Lab Manager

Signature:

Date Signed:

7/31/2015

Mailing Address: 200 N Lake Street

(enter only if different
from address above)

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City: Cadillac

State: MI

Postal Code: 49601 05 2015

Phone Number: (231) 775-2368

E-mail address: lab@cadillac-mi.net

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

United States
ENVIRONMENTAL PROTECTION AGENCY

Laboratory Performance Evaluation
Laboratory DMR-QA Evaluation Study 35

USEPA NPDES Permit #: MI0020257

Permit Ext:

Identification of all CHEM, MICRO and TOX laboratories who did analyses for this permit							
Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Lab**
			CHEM	MICRO	TOX		
Cadillac WWTP	200 N. Lake St. Cadillac, MI 49601	MI00082	☒	☒	☐	G	☒
Kar Labs	4425 Manchester Rd. Kalamazoo, MI 49001	MI00041	☒	☐	☐	C	☒
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C	☒

* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote 2 on DMRQA-35, Frequently Asked Questions page

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 35

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C				
Hardness Total Suspended Solids Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃				
pH pH				
Settleable Solids Settleable Solids				
Solids Concentrate Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Ammonia as N Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P				
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P				
Nitrite Nitrite as N				
Demand BOD CBOD COD TOC				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 35

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Oil & Grease Concentrate				
n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease				
n-Hexane Extractable Material(O&G) (Grav)				
n-Hexane Extractable Material(O&G) (IR)				
Trace Metals				
Aluminum				
Antimony				
Arsenic				
Barium				
Beryllium				
Boron				
Cadmium				
Chromium				
Cobalt				
Copper				
Iron				
Lead				
Manganese				
Molybdenum				
Nickel				
Selenium				
Silver				
Strontium				
Thallium				
Vanadium				
Zinc				
Mercury				
Mercury				
Low-Level Mercury				
Low Level Mercury EPA 1631E	☐	☐	☒	☐
Hexavalent Chromium				
Hexavalent Chromium				
Turbidity				
Turbidity				
Total Cyanide				
Cyanide, total				
Amenable Cyanide				
Total Phenolics (4-AAP)				
Phenolics, total				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 35

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Total Residual Chlorine				
Total Residual Chlorine				
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 35

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C				
Hardness Total Suspended Solids SM 2540 D-2011 Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃	☐	☒	☐	☐
pH pH SM 4500-H+ B-2011	☐	☒	☐	☐
Settleable Solids Settleable Solids				
Solids Concentrate Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Ammonia as N SM 4500-NH ₃ D-2011 Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P	☐	☒	☐	☐
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P SM 4500-P E-2011	☐	☒	☐	☐
Nitrite Nitrite as N				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 35

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Demand BOD CBOD SM 5210 B-2011 COD TOC	☐	☒	☐	☐
Oil & Grease Concentrate n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease n-Hexane Extractable Material(O&G) (Grav) n-Hexane Extractable Material(O&G) (IR)				
Trace Metals Aluminum Antimony Arsenic Barium Beryllium Boron Cadmium Chromium Cobalt Copper Iron Lead Manganese Molybdenum Nickel Selenium Silver Strontium Thallium Vanadium Zinc				
Mercury Mercury				
Low-Level Mercury Low Level Mercury				
Hexavalent Chromium Hexavalent Chromium				
Turbidity Turbidity				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist DMRQA Study 35

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
<u>Total Cyanide</u>				
Cyanide, total				
Amenable Cyanide				
<u>Total Phenolics (4-AAP)</u>				
Phenolics, total				
<u>Total Residual Chlorine</u>				
Total Residual Chlorine				
<u>Low-Level Total Residual Chlorine</u>				
Low Level Total Residual Chlorine				
<u>WasteWatR™ Coliform MicrobE™</u>				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
SM 9222 D (m-FC)-1997				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMRWet Study 35

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
Fathead minnow (Test Code 13)						
0754	48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2000.0	LC50	☐	☒	☐	☐
Fathead minnow (Test Code 14)						
0755	48Hr., Acute, Non-Renewal, 25° C, 20% DMW	LC50				
Fathead minnow (Test Code 15)						
0808	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	IC25 (ON) Growth	☐	☒	☐	☐
0810	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	NOEC (ON) Growth	☐	☒	☐	☐
0756	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	NOEC Survival	☐	☒	☐	☐
Fathead minnow (Test Code 16)						
0812	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 (ON) Growth				
0814	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC (ON) Growth				
0759	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
Ceriodaphnia dubia (Test Code 19)						
0764	48Hr., Acute Renewal, 25° C, MHSF EPA 2002	LC50	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 20)						
0765	48Hr., Acute Renewal, 25° C, 20% DMW	LC50				
Ceriodaphnia dubia (Test Code 21)						
0767	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	IC25 Reproduction	☐	☒	☐	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	NOEC Reproduction	☐	☒	☐	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	NOEC Survival	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 22)						
0770	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 Reproduction				
0771	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Reproduction				
0769	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
Daphnia magna (Test Code 32)						
0788	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50				
Daphnia pulex (Test Code 38)						
0794	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMRWet Study 35

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
<u>Mysid (Test Code 42)</u>						
0798	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Mysid (Test Code 43)</u>						
0816	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth				
0818	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth				
0799	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival				
<u>Inland silverside (Test Code 44)</u>						
0803	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Inland Silverside (Test Code 45)</u>						
0825	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth				
0826	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth				
0824	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival				
<u>Sheepshead minnow (Test Code 46)</u>						
0804	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Sheepshead minnow (Test Code 47)</u>						
0820	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth				
0822	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth				
0805	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival				

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used



DMR-QA 35 Corrective Action Guidelines

Permit Holders

Upon reviewing your enclosed DMR-QA 35 final report, you may notice that some analyte results have been evaluated as 'Not Acceptable'. For these analytes, the USEPA's DMR-QA 35 announcement letter directs permit holders to "consult with the laboratory and investigate any discrepancies or "Not Acceptable" evaluations reported by the PT Provider. Permittees must identify and report to the state DMR-QA coordinator causes and system changes to correct the discrepancies, and avoid their reoccurrence." The corrective action report must be sent to the coordinator before October 23, 2015. This means that for all 'Not Acceptable' evaluations, the laboratory must conduct an investigation, identify the problem and implement adequate corrective actions. Your state may also require the laboratory to analyze a remedial QC or PT Sample as a demonstration that the corrective action implemented was successful. ERA has contacted all state and regional coordinators and determined the type of remedial samples that they recommend or require.

If any of your 'Not Acceptable' results originated from your in-house laboratory, an order form is enclosed listing the standards that you should order and analyze. Your in-house laboratory will need to conduct its' investigation and then analyze the listed standards to ensure that the identified problem has been corrected. The corrective action report and, if applicable, the results of the analysis will then need to be forwarded to your DMR-QA coordinator by October 23, 2015.

You may receive an order form for both your in-house standards that have analytes evaluated as 'Not Acceptable', as well as a list of standards that your outside or contract laboratory(ies) have with 'Not Acceptable' evaluations. Some agencies require corrective action on a case by case basis. If you have a "Not Acceptable" evaluation, but there is not an order form or a list of standards for your in-house or outside laboratories, ERA recommends that you contact your DMR-QA Coordinator for their corrective action requirements, if any.

The standards listed in the attachments are based on general guidance given to us by the state or regional coordinators. If you have any questions about your specific corrective action requirements, please contact your state or regional coordinator directly.

If you have any questions or to place an order, please contact ERA's Customer Service Group at 1-888-372-0122.

CERTIFICATE OF RECOGNITION

*ERA congratulates
CADILLAC WWTP*

for your participation and successful evaluation in DMR-QA 35. We recognize the performance of this laboratory for achieving acceptable evaluations for the following analytes.

Ammonia as N
CBOD

Fecal Coliforms (MF)
pH

Total phosphorus as P
Total Suspended Solids



Kristina Sanchez
Quality Officer

C454401



A Waters Company

Study: **DMR-QA 35**
NPDES Permit #: **MI0020257**
Laboratory Name: **Cadillac WWTP**

Inorganic Results





A Waters Company

DMR-QA 35 Final Report

NPDES Permit #: M10020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/31/15
Study Dates: 03/20/15 - 07/10/15

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Hardness (cat# 580)											
1960	Total Suspended Solids	mg/L	Acceptable	57.9	71.7	57.9 - 80.3	SM 2540 D-2011 2011	65.9	4.58	MI00082	DMRQA35
DMRQA pH (cat# 577)											
1900	pH	S.U.	Acceptable	6.42	6.43	6.23 - 6.63	SM 4500-H+ B-2011 2011	6.41	0.0572	MI00082	DMRQA35
DMRQA Simple Nutrients (cat# 584)											
1515	Ammonia as N	mg/L	Acceptable	9.84	10.2	8.12 - 12.2	SM 4500-NH3 D-2011 2011	10.2	0.770	MI00082	DMRQA35
DMRQA Complex Nutrients (cat# 579)											
1910	Total phosphorus as P	mg/L	Acceptable	3.77	3.34	2.74 - 3.91	SM 4500-P E-2011 2011	3.45	0.193	MI00082	DMRQA35
DMRQA Demand (cat# 578)											
1555	CBOD	mg/L	Acceptable	47.3	41.5	18.1 - 64.9	SM 5210 B-2011 2011	44.3	9.83	MI00082	DMRQA35
DMRQA Low-Level Mercury (cat# 896)											
1095	Low Level Mercury	ng/L	Not Acceptable	86	56.8	41.4 - 71.6	EPA 1631E 2002	55.5	4.52	MI00041	DMRQA35



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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Study #: DMR-QA 35

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A Waters Company

Study: **DMR-QA 35**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Microbiology Results



Study #: DMR-QA 35

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A Waters Company

DMR-QA 35 Final Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Pielt Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/31/15
Study Dates: 03/20/15 - 07/10/15

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
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DMRQA WasteWatR™ Coliform MicrobE™ (cat# 576)

2530	Fecal Coliforms (MF)	CFU/100mL	Acceptable	68	354	4.00 - 1010	SM 9222 D (m-FC)-1997	66.7	167	MI00082	DMRQA35
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Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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Study #: DMR-QA 35

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A Waters Company

Study: **DMR-QA 35**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

WET Results



Study #: DMR-QA 35

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DMR-QA 35 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/31/15
Study Dates: 03/20/15 - 07/10/15

Analyte Code	Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRWet Fathead minnow (Test Code 13) (cat# WET002) 48Hr., Acute, Non-Renewal, 25° C, MHSF Potassium chloride										
0754	LC50	Acceptable	62.1	40.7	17.8 - 63.6	EPA 2000.0 2002	40.7	11.5	MI01187	DMRWet35
DMRWet Fathead minnow (Test Code 15) (cat# WET004) 7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0808	IC25 (ON) Growth	Acceptable	32.2	33.1	18.3 - 47.8	EPA 1000 2002	33.1	7.37	MI01187	DMRWet35
0810	NOEC (ON) Growth	Acceptable	25	25.0	12.5 - 50.0	EPA 1000 2002			MI01187	DMRWet35
0756	NOEC Survival	Acceptable	25	25.0	12.5 - 50.0	EPA 1000 2002			MI01187	DMRWet35
DMRWet Ceriodaphnia dubia (Test Code 19) (cat# WET008) 48Hr., Acute Renewal, 25° C, MHSF Potassium chloride										
0764	LC50	Acceptable	60.0	54.9	26.5 - 83.3	EPA 2002	54.9	14.2	MI01187	DMRWet35
DMRWet Ceriodaphnia dubia (Test Code 21) (cat# WET010) 7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0767	IC25 Reproduction	Acceptable	25.3	45.6	19.2 - 72.0	EPA 1002 2002	45.6	13.2	MI01187	DMRWet35
0768	NOEC Reproduction	Acceptable	50	25.0	12.5 - 50.0	EPA 1002 2002			MI01187	DMRWet35
0766	NOEC Survival	Acceptable	50	50.0	25.0 - 100	EPA 1002 2002			MI01187	DMRWet35



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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Study #: DMR-QA 35

Page 6 of 6



200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

September 10, 2015

Dear Chris Veldkamp,

I am the Lab Manager for the City of Cadillac WWTP (MI 0020257). We use Kar Laboratories for our low level mercury sampling. During the DMR-QA 35 testing process they received an "Unacceptable" on their low level mercury PT sample. I have received their corrective action for this analysis. I have also spoken to Dave Alkema (Kar Lab Manager) about the reason behind their deficiency and how they addressed it. I have included their corrective action letter and "Acceptable" result on an additional sample.

If you have any questions or comments please feel free to contact me by email at lab@cadillac-mi.net or telephone at (231)775-2368.

Thank you,

A handwritten signature in black ink, appearing to read "Amy Vail", is written over the printed name.

Amy Vail
City of Cadillac
Lab Manager

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SEP 14 2015

**DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS**

4425 Manchester Road

Kalamazoo, MI 49001

Phone 269 381-9666

Fax 269 381-9698

www.karlabs.com

August 18, 2015

City of Cadillac, Water Resources
Attn: Ms. Amy Vail
200 North Lake Street
Cadillac, MI 49601

Dear Ms. Vail,

This letter represents our Corrective Action Report for the unacceptable Low-Level Mercury result we recently reported for DMR-QA Study 35. The Study 35 test sample was prepared from a concentrate following the supplier's instructions, and KAR reported a result of 86 ng/L, which fell outside of the 41.4 – 71.6 ng/L acceptance range.

Our initial investigation uncovered no obvious error in the data. All QC measures indicated that our calibration was accurate and the test was in control. However, while analyzing a remedial sample from the supplier, we were able to determine that the laboratory glassware used to prepare the original test sample was contaminated with mercury. KAR has since reported an acceptable result on the remedial sample, and we have dedicated clean glassware exclusively to the preparation of Low-level Mercury performance study samples.

We apologize for the inconvenience caused by this incident, and thank you for your patience. Because these blind-study samples must be prepared from a concentrate, they are exposed to potential sources of contamination that client samples are not. **The accuracy of our Low-level Mercury results for client samples was not affected by this failure. The contamination problem we encountered involved only the test sample prepared from the supplier's concentrate.**

Sincerely,



David R. Alkema
Laboratory Manager

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SEP 14 2015

**DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS**



080515N 2009 TNI Evaluation Final Complete Report

A Waters Company

William G. Rauch
President
Kar Labs
4425 Manchester Rd.
Kalamazoo, MI 49001
269-381-9666

EPA ID:
ERA Customer Number:

MI00041
K060501

TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
MP Low-Level Mercury (cat# 931) Study Dates: 08/05/15 - 08/12/15												
1095	Low Level Mercury	ng/L	54.3	64.4	47.9 - 80.1	Acceptable	EPA 1631E 2002	8/12/2015		65.2	0.00	



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
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